

D6_Dels_Transformers_als_LLM

March 2, 2026

1 D6.Dels Transformers als LLM

1.1 El problema dels embeddings estàtics

Word2Vec va representar un gran avenç en el processament del llenguatge natural. Gràcies a aquest model vam aprendre a representar paraules com vectors numèrics amb significat matemàtic, cosa que va permetre als ordinadors captar relacions com similitud, analogia o proximitat de significat.

Per primera vegada, les màquines van començar a entendre parcialment el contingut d'un text, ja que podien treballar amb representacions que reflectien el significat de les paraules.

En el notebook anterior vam veure que en Word2Vec cada paraula té un únic vector fix. Aquesta és una limitació important, perquè el llenguatge és inherentment ambigu i contextual: la mateixa paraula pot variar molt segons la frase on apareix. Per exemple, "banc" pot significar institució financera, moble del parc, banc de peixos o banc de dades.

En un model com Word2Vec, el vector final de banc és aproximadament la mitjana de tots els contexts on la paraula apareix:

```
vec("banc") = mitjana(  
    "institució financera"  60% usos  
    "moble del parc"       25% usos  
    "banc de peixos"       10% usos  
    "banc de dades"        5% usos  
)
```

Això vol dir que el vector de banc acaba sent una barreja de tots els significats possibles, sense distingir en quin sentit s'utilitza en cada cas concret. Aquesta rigidesa fa que els embeddings estàtics tinguin dificultats per captar el context real d'una paraula

1.2 ELMo (2018): el primer pas cap al context

ELMo (*Embeddings from Language Models*) va ser el primer sistema a generar **vectors que canvien** dependent del context. Va guanyar el premi al millor article de l'NAACL 2018.

ARQUITECTURA ELMo (Peters et al., 2018)

Input: "I went to the [bank] to deposit money"

↓

Capa 1 (caràcters → subparaules)

"bank" → b-a-n-k → embedding morfològic

↓
 Capa 2 (LSTM → esquerra)
 "I went to the" → context esquerra de "bank"
 ↓
 Capa 3 (LSTM → dreta)
 "to deposit money" → context dreta de "bank"
 ↓
 COMBINA les 3 capes → vec_bank_FINANCER

Input: "I sat on the [bank] by the river"
 ... (mateix procés) ...
 → vec_bank_GEOGRAFIC ← VECTOR DIFERENT!

ELMo genera un vector diferent per cada aparició
 de la mateixa paraula, depenent de les paraules del voltant.

```
[1]: !pip install sentence-transformers gensim
```

```

Requirement already satisfied: sentence-transformers in
/usr/local/lib/python3.12/dist-packages (5.2.3)
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (5.0.0)
Requirement already satisfied: huggingface-hub>=0.20.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.4.1)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-
packages (from sentence-transformers) (2.10.0+cpu)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-
packages (from sentence-transformers) (1.6.1)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.15.0)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (4.67.3)
Requirement already satisfied: smart_open>=1.8.1 in
/usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.24.3)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-

```

transformers) (2025.3.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (1.3.0)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (0.28.1)

Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (26.0)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (6.0.3)

Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (1.5.4)

Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (0.24.0)

Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.1)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.6.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.6)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (2025.11.3)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (0.22.2)

Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (0.7.0)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-transformers) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-transformers) (3.6.0)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (4.12.1)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (2026.1.4)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers)

```

(1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
(from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-
hub>=0.20.0->sentence-transformers) (0.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from
sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sentence-
transformers) (3.0.3)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-
packages (from typer-slim->huggingface-hub>=0.20.0->sentence-transformers)
(0.24.1)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-
packages (from typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-
packages (from typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->huggingface-hub>=0.20.0->sentence-transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from
rich>=12.3.0->typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from
rich>=12.3.0->typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-
slim->huggingface-hub>=0.20.0->sentence-transformers) (0.1.2)
Downloading
gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9
MB)
27.9/27.9 MB
40.5 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0

```

```

[2]: import numpy as np
      from scipy.spatial.distance import cosine
      from sentence_transformers import SentenceTransformer
      import gensim.downloader as api

```

```

# GloVe estàtic (1 vector per "bank")
glove = api.load('glove-wiki-gigaword-100')
vec_bank_glove = glove['bank']

# ELMo-like contextual: model petit i ràpid (all-MiniLM-L6-v2 ~ BERT contextual)
model_ctx = SentenceTransformer('all-MiniLM-L6-v2')

frases_bank = [
    "I went to the bank to deposit money.", # financer
    "We sat on the grassy bank by the river.", # geogràfic
]

print("\nRESULTATS de la comparativa")

# GloVe: mateix vector sempre
sim_glove = 1 - cosine(vec_bank_glove, vec_bank_glove) # Sempre 1.0!
print(f"GloVe 'bank' vs 'bank': {sim_glove:.3f} (estàtic)")

# Contextual: embeddings de FRASE, però aprox token amb mean pooling simple
embeds_ctx = model_ctx.encode(frases_bank)
sim_ctx = 1 - cosine(embeds_ctx[0], embeds_ctx[1])
print(f"ELMo-like financer vs geogràfic: {sim_ctx:.3f} (contextual!!)")

```

```
[=====] 100.0% 128.1/128.1MB
downloaded
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.
```

```
warnings.warn(
```

```
modules.json: 0%| | 0.00/349 [00:00<?, ?B/s]
```

```
config_sentence_transformers.json: 0%| | 0.00/116 [00:00<?, ?B/s]
```

```
README.md: 0.00B [00:00, ?B/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
```

```
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.
```

```
sentence_bert_config.json: 0%| | 0.00/53.0 [00:00<?, ?B/s]
```

```

config.json: 0%|          | 0.00/612 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/90.9M [00:00<?, ?B/s]
Loading weights: 0%|          | 0/103 [00:00<?, ?it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                | Status          | |
-----+-----+---
embeddings.position_ids | UNEXPECTED    | |

Notes:
- UNEXPECTED      :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

tokenizer_config.json: 0%|          | 0.00/350 [00:00<?, ?B/s]
vocab.txt: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
special_tokens_map.json: 0%|          | 0.00/112 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/190 [00:00<?, ?B/s]

```

RESULTATS de la comparativa

GloVe 'bank' vs 'bank': 1.000 (estàtic)

ELMo-like financier vs geogràfic: 0.434 (contextual!!)

Com podem veure, Glove no captura el context però el model similar a ELMo si.

1.3 Els Transformers: l'arquitectura que ho ha canviat tot (el “boom” dels LLM)

“Attention is All You Need” — Google, 2017

És l'article més citat de la història del Deep Learning.

Fa un pas més en el camp de la NLP eliminant les LSTM i les RNN, i va proposar un mecanisme radicalment nou: **l'Atenció**.

Tots els models moderns (BERT, GPT, Claude, Gemini, LLaMA) en deriven directament.

Les LSTM (Long Short-Term Memory) i RNN (Recurrent Neural Networks) són models de xarxes neuronals dissenyats per processar seqüències de dades com text o sèries temporals, però tenen un problema fonamental de processament seqüencial: per entendre que “és” es refereix a “gat” en una frase com “El gat que es va asseure sobre l'estora és molt pèlut”, el model ha de processar tots els tokens un per un, cosa que les fa molt lentes per no poder-se paral·lelitzar en GPU, provoca gradients que s'esvaeixen en seqüències llargues i dificulta capturar dependències llunyanes.

La solució revolucionària fou el mecanisme d'atenció, que permet a cada token “mirar” directament tots els altres de forma simultània: “és” assigna un 85% d'atenció a “gat” i un 12% a “pelut” sense necessitar processar res seqüencialment, fent-ho ràpid, global i independent de la distància dins la frase.

VIDEO ¿Qué es un TRANSFORMER? La Red Neuronal que lo cambió TODO!
<https://www.youtube.com/watch?v=aL-EmKuB078>

1.4 Models pre-entrenats: BERT i GPT

El paradigma revolucionari dels models pre-entrenats com BERT i GPT (2018+) consisteix en dues fases:

- Pre-entrenament massiu en centenars de gigabytes de text (Wikipedia + llibres) durant mesos amb milers de GPUs, on models com BERT (bidireccional, 340M paràmetres) o GPT-3 (unidireccional, 175B paràmetres) aprenen les estructures lingüístiques completes d'una llengua.
- Fine-tuning ràpid (només 2-3 hores en 1 GPU) adapta aquest coneixement previ a la teva tasca específica amb poques dades (100-10k mostres), aconseguint rendiments d'estat de l'art que abans requerien setmanes d'entrenament des de zero i grans conjunts de dades.

##GPT: NLP Generatiu

GPT (*Generative Pre-trained Transformer*, OpenAI, 2018) va prendre el Transformer decoder i el va escalar.

La idea darrera de l'entrenament de GPT és molt simple: se li dona una quantitat enorme de text i se li demana sempre la mateixa cosa, predir el següent token de la seqüència i així repetit per centenars de milers de milions de tokens a escala d'internet.

```
[3]: from transformers import pipeline

# GPT-2 en CATALÀ
generator = pipeline('text-generation', model='Marxx01/gpt2_catalan')

resultats = generator(
    "L'IA del futur",
    max_new_tokens=20,
    num_return_sequences=3
)

for i, text in enumerate(resultats, 1):
    print(f"{i}. {text['generated_text']}")
```

```
config.json: 0%|          | 0.00/782 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/1.43G [00:00<?, ?B/s]
Loading weights: 0%|          | 0/292 [00:00<?, ?it/s]
generation_config.json: 0%|          | 0.00/132 [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/440 [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
merges.txt: 0.00B [00:00, ?B/s]
special_tokens_map.json: 0%|          | 0.00/99.0 [00:00<?, ?B/s]
```

tokenizer.json: 0.00B [00:00, ?B/s]

Passing `generation_config` together with generation-related arguments like `num_return_sequences` or `max_new_tokens` is deprecated and will be removed in future versions. Please pass either a `generation_config` object OR all generation parameters explicitly, but not both.

Both `max_new_tokens` (=20) and `max_length` (=20) seem to have been set.

`max_new_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

1. L'IA del futur de la intel·ligència artificial i la robòtica

L'IA del futur de la intel·

2. L'IA del futur

El Govern espanyol ha decidit que el preu de l'energia es rebaixi al 50%

3. L'IA del futur no serà el nou Ikea, sinó l'Ikea de sempre

Els treballadors de la

1.5 Hugging Face: NLP industrial accessible

Hugging Face és el “GitHub dels models d’IA”: +500.000 models, +100.000 datasets, tots descarregables en 2 línies de codi. Ha democratitzat l’accés als models estat-de-l’art.

ECOSISTEMA HUGGING FACE

Hub	: models, datasets, spaces (demos)
Transformers	: descarregar i usar qualsevol model
Datasets	: carregar datasets estàndard
Evaluate	: mètriques (BLEU, F1, accuracy...)
PEFT	: fine-tuning eficient (LoRA, QLoRA...)
Spaces	: deploy de demos amb Gradio/Streamlit

```
[4]: #HUGGING FACE - disposa de pipelines per a múltiples tasques NLP
from transformers import pipeline

#sentiment analysis
sentiment = pipeline("sentiment-analysis",
                     model="distilbert-base-uncased-finetuned-sst-2-english")

frases_sent = [
    "This new Python library is absolutely fantastic and easy to use!",
    "The installation process was a complete disaster. Nothing works.",
    "The documentation is okay, but could be improved.",
    "I've been using this framework for 3 years and it's still great.",
```



```

]
for f in frases_sent:
    r = sentiment(f)[0]
    print(f"[{r['label']:8}] ({r['score']:.3f})")
    print(f"      '{f[:60]}...'")

# NER (Named Entity Recognition)
ner = pipeline("ner", model="dslim/bert-base-NER",
               aggregation_strategy="simple")

text_ner = "Elon Musk founded Tesla in Palo Alto, California in 2003 with $6.5M_
↳funding."
entitats = ner(text_ner)
print(f"  Text: '{text_ner}'")
for e in entitats:
    print(f"[{e['entity_group']:4}] '{e['word']}' ({e['score']:.3f})")

# Q&A
qa = pipeline("question-answering",
              model="deepset/roberta-base-squad2")

context = """
BERT (Bidirectional Encoder Representations from Transformers) was developed
by researchers at Google AI Language and published in 2018. It was pre-trained
on English Wikipedia and BooksCorpus with 3.3 billion words. BERT achieved
state-of-the-art results on 11 NLP tasks simultaneously.
"""

questions = [
    "Who developed BERT?",
    "When was BERT published?",
    "How many NLP tasks did BERT improve?",
]

for q in questions:
    r = qa(question=q, context=context)
    print(f"{q}")
    print(f"'{r['answer']}' (confiança: {r['score']:.3f})")
    print()

```

```

config.json:  0%|          | 0.00/629 [00:00<?, ?B/s]
model.safetensors:  0%|          | 0.00/268M [00:00<?, ?B/s]
Loading weights:  0%|          | 0/104 [00:00<?, ?it/s]
tokenizer_config.json:  0%|          | 0.00/48.0 [00:00<?, ?B/s]
vocab.txt:  0%|          | 0.00/232k [00:00<?, ?B/s]

```

[POSITIVE] (1.000)
 'This new Python library is absolutely fantastic and easy to ...'
 [NEGATIVE] (1.000)
 'The installation process was a complete disaster. Nothing wo...'
 [NEGATIVE] (0.943)
 'The documentation is okay, but could be improved...'
 [POSITIVE] (1.000)
 'I've been using this framework for 3 years and it's still gr...'

config.json: 0%| | 0.00/829 [00:00<?, ?B/s]
 model.safetensors: 0%| | 0.00/433M [00:00<?, ?B/s]
 Loading weights: 0%| | 0/199 [00:00<?, ?it/s]

BertForTokenClassification LOAD REPORT from: dslim/bert-base-NER

Key	Status		
bert.pooler.dense.bias	UNEXPECTED		
bert.pooler.dense.weight	UNEXPECTED		

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture;
 not ok if you expect identical arch.

tokenizer_config.json: 0%| | 0.00/59.0 [00:00<?, ?B/s]
 vocab.txt: 0.00B [00:00, ?B/s]
 added_tokens.json: 0%| | 0.00/2.00 [00:00<?, ?B/s]
 special_tokens_map.json: 0%| | 0.00/112 [00:00<?, ?B/s]

Text: 'Elon Musk founded Tesla in Palo Alto, California in 2003 with \$6.5M funding.'

[ORG] 'Elon Musk' (0.992)
 [ORG] 'Tesla' (0.997)
 [LOC] 'Palo Alto' (0.994)
 [LOC] 'California' (0.999)

config.json: 0%| | 0.00/571 [00:00<?, ?B/s]
 model.safetensors: 0%| | 0.00/496M [00:00<?, ?B/s]
 Loading weights: 0%| | 0/199 [00:00<?, ?it/s]

RobertaForQuestionAnswering LOAD REPORT from: deepset/roberta-base-squad2

Key	Status		
roberta.embeddings.position_ids	UNEXPECTED		

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture;
 not ok if you expect identical arch.

```
tokenizer_config.json: 0%|          | 0.00/79.0 [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
merges.txt: 0.00B [00:00, ?B/s]
special_tokens_map.json: 0%|          | 0.00/772 [00:00<?, ?B/s]
Who developed BERT?
'Google AI Language' (confiança: 0.764)

When was BERT published?
'2018' (confiança: 0.967)

How many NLP tasks did BERT improve?
'11' (confiança: 0.868)
```

1.6 LLMs: Models de Llengua Grans

Alguns exemples: GPT-3 (2020), PaLM (2022), LLaMA (2023), Claude (2023), GPT-4 (2023), Gemini (2024).

Escalar els Transformers fins a centenars de milers de milions de paràmetres ha donat lloc a capacitats emergents que ningú havia planificat: models que “raonen”.

L’arquitectura és senzilla però només cal mirar el temps i cost d’entrenament:

Model	Params	Any	Cost entrenament
GPT-2	1.5B	2019	\$50.000
GPT-3	175B	2020	\$12M
PaLM	540B	2022	\$9M
GPT-4	~1T	2023	~\$100M
LLaMA-3	70B	2024	open source
Claude 3	??B	2024	Anthropic

Prompt Engineering és la ciència/tècnica que permet dissenyar instruccions precises en llenguatge natural per obtenir els millors resultats dels Large Language Models (LLMs) sense necessitar entrenament addicional. És una nova forma de “programació” on controles el comportament del model només amb el text del prompt.

Principals tècniques:

- **Zero-shot:** dones la instrucció directament sense exemples. Útil per a tasques simples i clares que el model ja coneix del seu entrenament.
- **Few-shot:** donar 2-3 exemples perquè copïi el patró. Útil quan necessites un format específic de resposta.
- **Chain-of-Thought (CoT):** li demanes “pensar pas a pas” per raonaments complexos. Utilitzar en problemes matemàtics, de lògica, etc.

- **System Prompt:** definir un “personatge” o rol del model quan necessitem un estil, to, rol específic (professor, advocat, etc.).

1.7 RAG: Connectar LLMs amb Coneixement Extern

RAG (Retrieval-Augmented Generation) resol el gran problema dels LLMs: només saben el que van aprendre fins a una data fixa (ex: GPT-4 coneix fins abril 2023). No saben preus actuals, documents de la teva empresa ni lleis noves, i poden inventar coses falses (al·lucinacions).

RAG funciona en 4 passos senzills:

- Preparació: Els teus documents (PDFs, bases de dades, preus...) es divideixen en fragments i s'indexen.
- Cerca intel·ligent: La pregunta de l'usuari s'analitza i es troben els fragments més rellevants.
- Prompt controlat: Es genera automàticament: “Basant-te EXCLUSIVAMENT en aquest context: [fragments]. Respon”.
- Generació segura: L'LLM produeix respostes verificables sense invencions.

Resultat: Accés en temps real a informació actualitzada i privada, eliminant al·lucinacions. És un LLM amb memòria externa personalitzable.

```
[5]: !pip install sentence-transformers scikit-learn
```

```
Requirement already satisfied: sentence-transformers in
/usr/local/lib/python3.12/dist-packages (5.2.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-
packages (1.6.1)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (5.0.0)
Requirement already satisfied: huggingface-hub>=0.20.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.4.1)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-
packages (from sentence-transformers) (2.10.0+cpu)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.15.0)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from sentence-transformers) (4.67.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.24.3)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-
transformers) (2025.3.0)
```

Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (1.3.0)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (0.28.1)

Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (26.0)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (6.0.3)

Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (1.5.4)

Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (0.24.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.6.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.6)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (2025.11.3)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (0.22.2)

Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->sentence-transformers) (0.7.0)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (4.12.1)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (2026.1.4)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (1.0.9)

Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (3.11)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers) (0.16.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from

sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
 Requirement already satisfied: MarkupSafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sentence-
 transformers) (3.0.3)
 Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-
 packages (from typer-slim->huggingface-hub>=0.20.0->sentence-transformers)
 (0.24.1)
 Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-
 packages (from typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
 transformers) (8.3.1)
 Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-
 packages (from typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
 transformers) (13.9.4)
 Requirement already satisfied: annotated-doc>=0.0.2 in
 /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
 slim->huggingface-hub>=0.20.0->sentence-transformers) (0.0.4)
 Requirement already satisfied: markdown-it-py>=2.2.0 in
 /usr/local/lib/python3.12/dist-packages (from
 rich>=12.3.0->typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
 transformers) (4.0.0)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
 /usr/local/lib/python3.12/dist-packages (from
 rich>=12.3.0->typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
 transformers) (2.19.2)
 Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
 packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-
 slim->huggingface-hub>=0.20.0->sentence-transformers) (0.1.2)

```

[6]: from sentence_transformers import SentenceTransformer
    from sklearn.metrics.pairwise import cosine_similarity
    import numpy as np

documents = [
    "AVUI 27/02/2026 YELMO BARCELONA: Dune 3 (17:30, 20:15, 22:45)",
    "AVUI 27/02/2026 YELMO BARCELONA: Oppenheimer 2 (18:00, 21:00)",
    "Preus Yelmo: Adult 10.5€ dilluns-jueves, 12€ caps de setmana"
]

model = SentenceTransformer('all-MiniLM-L6-v2')
doc_embs = model.encode(documents)

query = "A quina hora passa Dune avui?"
query_emb = model.encode([query])

sims = cosine_similarity(query_emb, doc_embs)[0]
top_docs = [documents[i] for i in sims.argsort()[::-1][:2]]
  
```

```

print(f"{query}")
print("RAG troba:")
for i, doc in enumerate(top_docs, 1):
    print(f"  {i}. {doc}")

```

Loading weights: 0%| | 0/103 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status		
-----+-----+--+-			
embeddings.position_ids	UNEXPECTED		

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

A quina hora passa Dune avui?

RAG troba:

1. AVUI 27/02/2026 YELMO BARCELONA: Dune 3 (17:30, 20:15, 22:45)
2. Preus Yelmo: Adult 10.5€ dilluns-jueves, 12€ caps de setmana